

Generating a Multimodal AI Framework for High Resolution Analysis of scATAC-seq Data on Human Immunity

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Introduction: Why is scATAC-seq Analysis Challenging?

Single cell Assay for Transposase-Accessible Chromatin Sequencing (scATAC-seq) enables unbiased identification of epigenetic activity at single cell-scale, but suffers from major challenges in cell type identification and annotation.

Challenges: No ground truth cell type feature set & difficulties in feature selection

- Unlike RNA, cell type regions are undefined for ATAC-seq.
- Features defined using CellRanger or other peak callers from bulk ATAC-seq will miss markers for small populations
- Low sample number in existing human PBMC scATAC-seq

Solution: Utilize the trimodal Transcripts, Epitopes, & Accessibility Sequencing (TEA-seq^{4,5}) data to define ground truth cell labels for scATAC-seq data

- Define cell type labels utilizing RNA & ADTs
- Perform feature selection by cell type using MOCHA¹
- Apply variational autoencoders (VAE) for label transfer & co-embedding to build atlas.

Finally, we apply our model to label and co-embed a scATAC-seq dataset of >3 million cells from 206 samples.

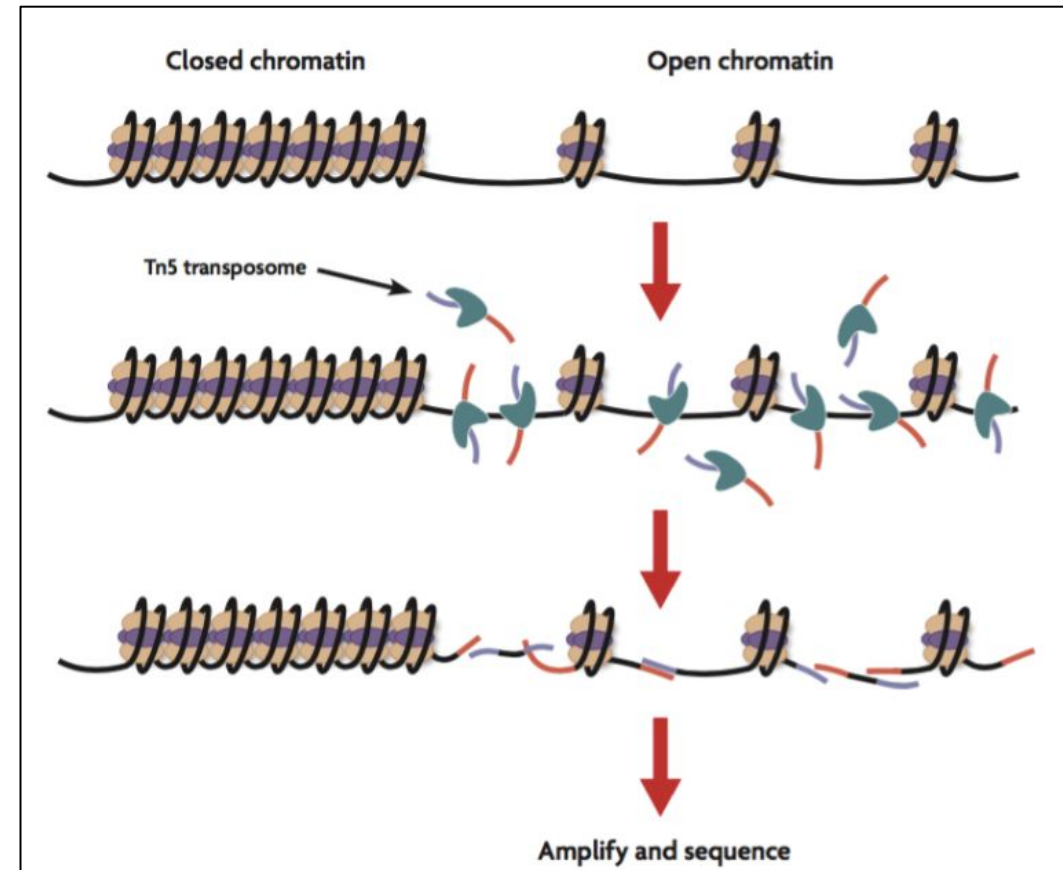


Figure 1: Image from ActiveMotif.com, reflecting how scATAC-seq protocol identifies open chromatin. <https://www.activemotif.com/blog-atac-seq>

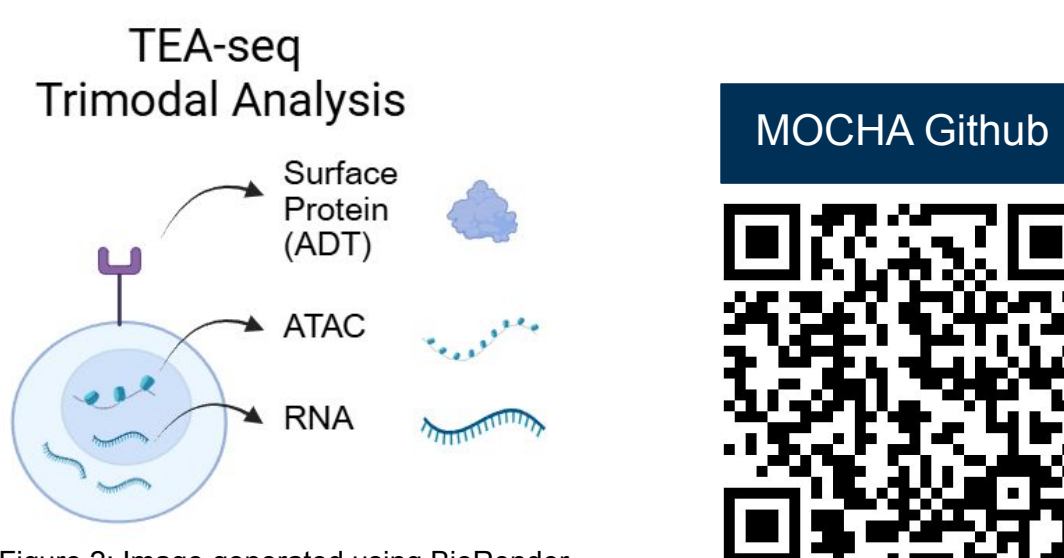


Figure 2: Image generated using BioRender

Solution: Trimodal Data for Feature Selection and AI Training

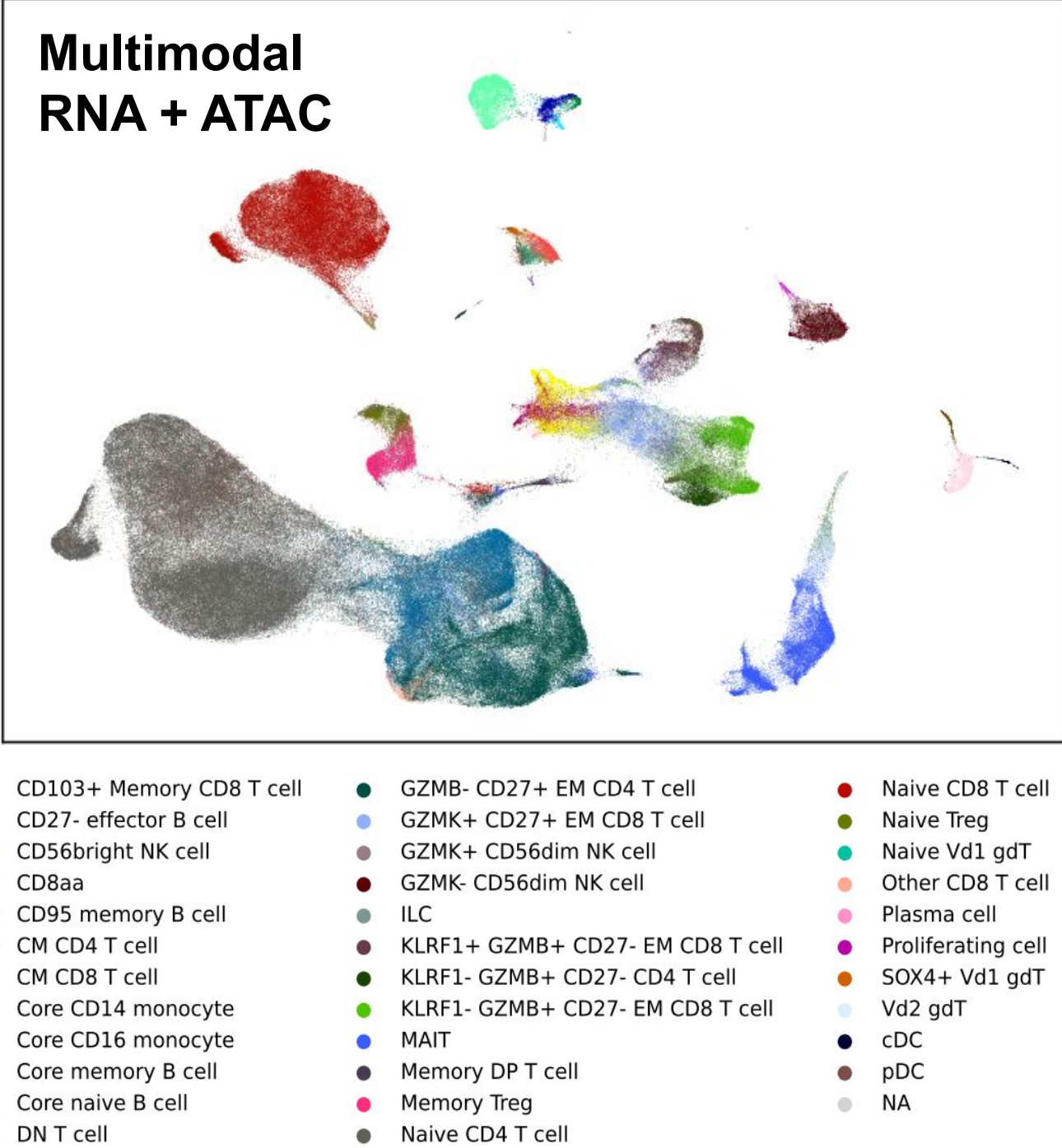
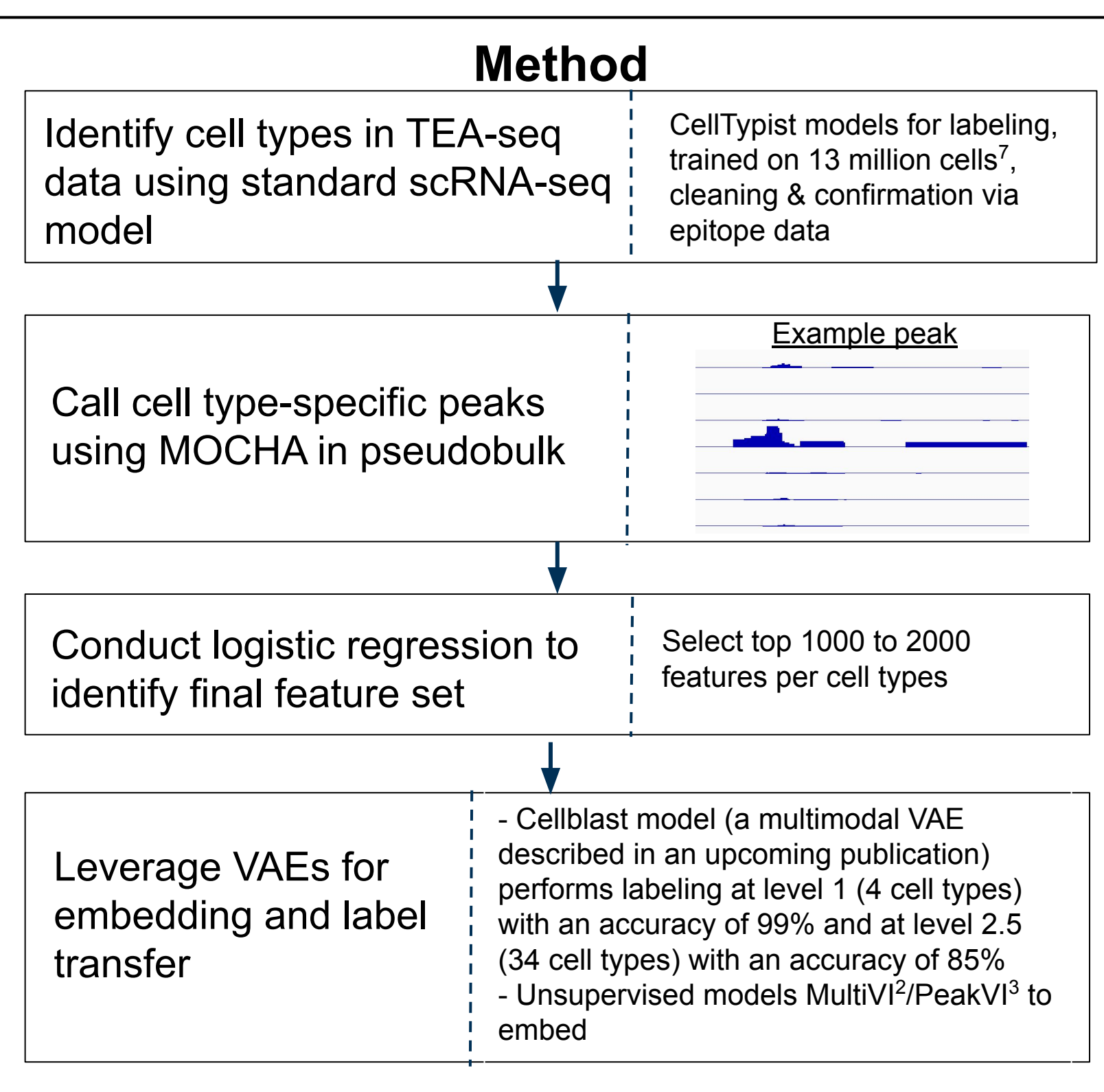


Figure 3. MultiVI UMAP embedding of final TEA-seq training dataset of 400k cells with 34 immune cell types.

Application: 3 Million Cell scATAC-seq Immune Health Atlas

Our peakset and model are generalizable to scATAC-seq datasets.

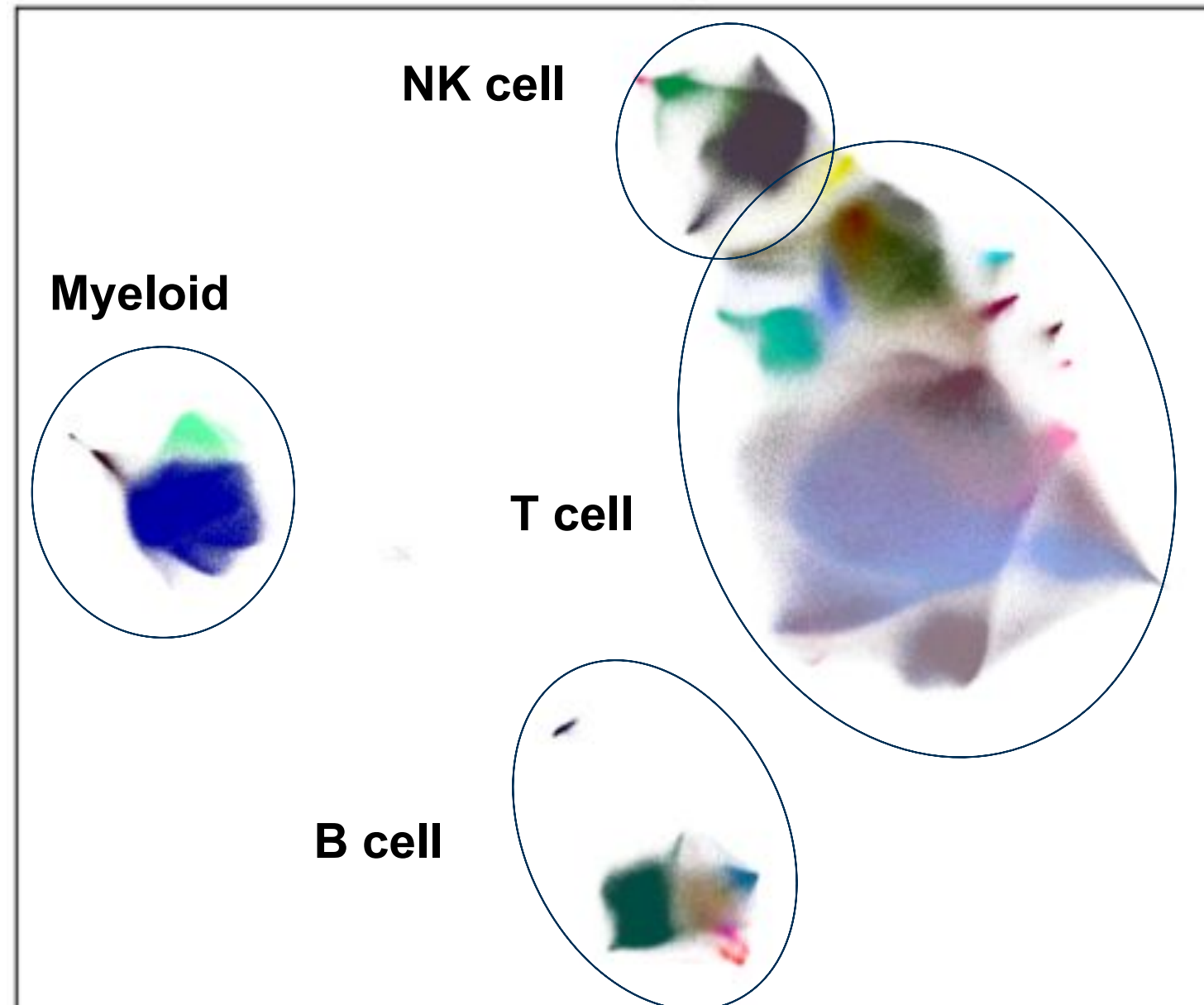


Figure 6. PeakVI UMAP embedding of our 3 million cell scATAC-seq reference contains 206 longitudinal samples from 78 healthy human donors to create an epigenetic atlas of the peripheral immune system across age and flu vaccine response timepoints.

- Adaptive NK cell
- CD8aa
- CD27+ effector B cell
- CD27- effector B cell
- CD56bright NK cell
- CD95 memory B cell
- CD103+ Memory CD8 T cell
- CM CD4 T cell
- CM CD8 T cell
- Core CD14 monocyte
- Core CD16 monocyte
- Core memory B cell
- Core naive B cell
- Core naive CD4 T cell
- Core naive CD8 T cell
- DN T cell
- GZMB+ Vd2 gdT
- GZMB- CD27+ EM CD4 T cell
- GZMK+ CD27+ EM CD8 T cell
- GZMK+ CD56 dim NK cell
- GZMK+ Vd2 gdT
- GZMK- CD56dim
- ILC
- KLR1+ GZMB+ CD27- EM CD8 T cell
- KLR1- GZMB+ CD27- CD4 T cell
- KLR1- GZMB+ CD27- EM CD8 T cell
- MAIT
- Memory DP T cell
- Memory Treg
- Naive Vd1 gdT
- Other CD8 T cell
- Plasma cell
- Proliferating cell
- SOX4+ Vd1 gdT
- cDC
- pDC

Co-embed paired sample datasets for cell type validation and recovery.

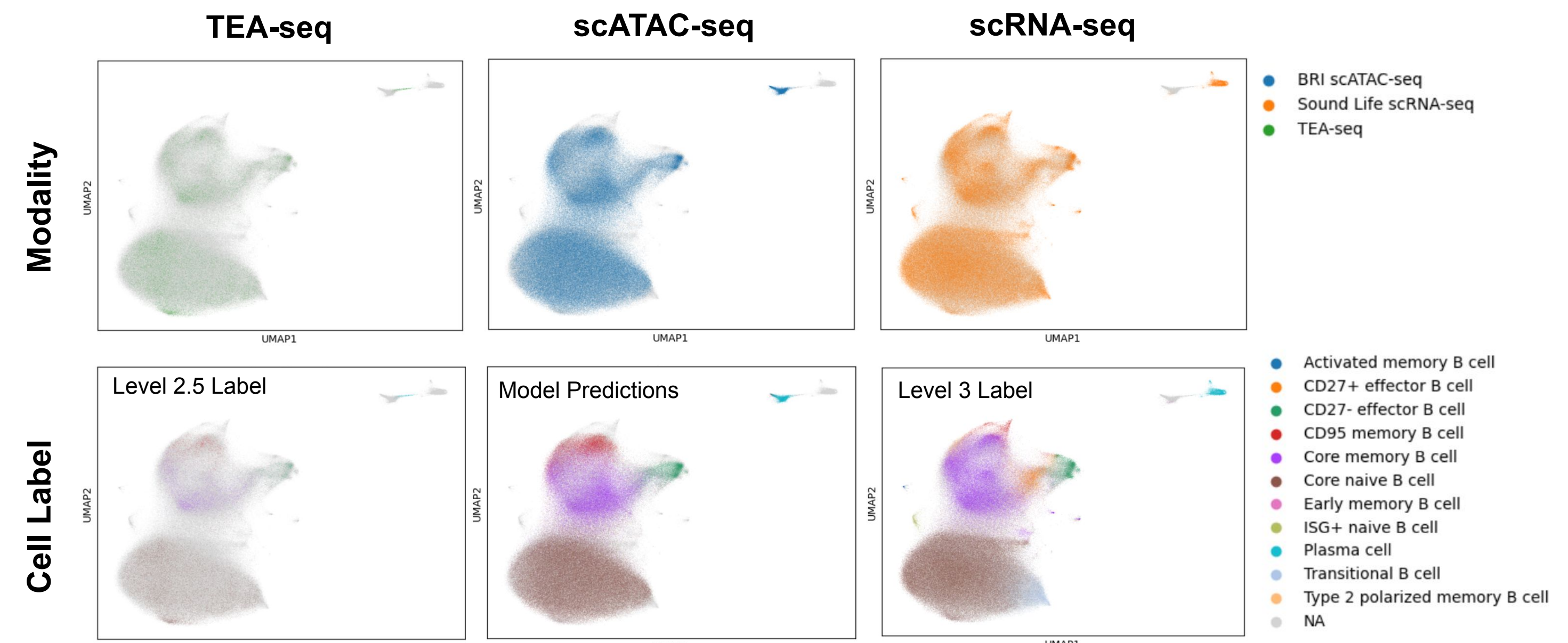
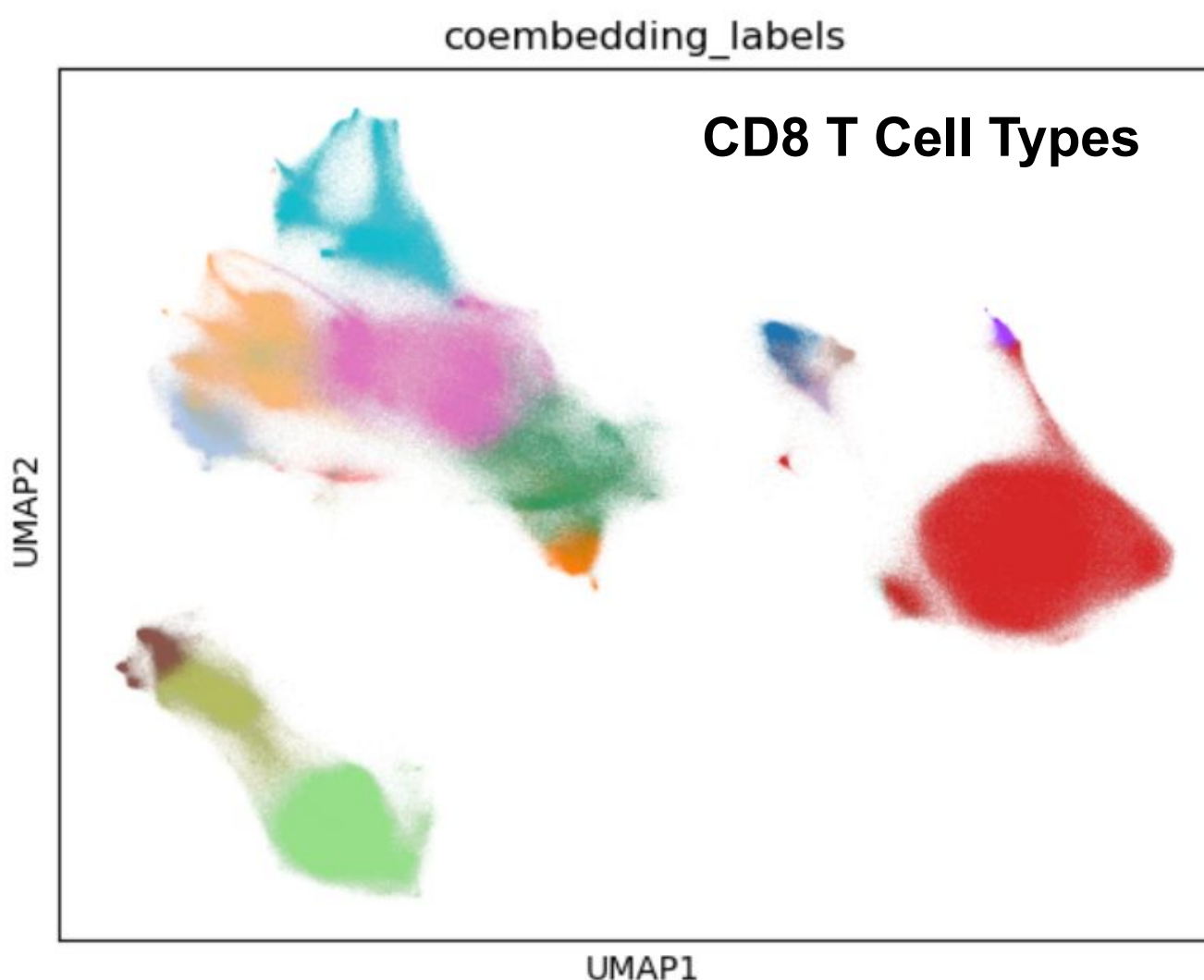


Figure 7. Cellblast model embeddings of paired samples of scRNA-seq (from the Sound Life cohort⁷) and scATAC-seq with our TEA-seq training data. Figure above shows UMAPs of B cells co-embedded across modality with predicted scATAC-seq labels validated by scRNA-seq labels.



Co-embedding within each major cell type enables the resolution of rare cell types.

Figure 8. Cellblast co-embedding shows high resolution labels of CD8 T cells.

- CD8aa
- CD103+ Memory CD8 T cell
- CM CD8 T cell
- Core naive CD8 T cell
- DN T cell
- GZMB+ Vd2 gdT
- GZMK+ CD27+ EM CD8 T cell
- KLR1+ GZMB+ CD27- EM CD8 T cell
- KLR1- GZMB+ CD27- CD4 T cell
- KLR1- GZMB+ CD27- EM CD8 T cell
- MAIT
- Memory DP T cell
- Naive Vd1 gdT
- SOX4+ Vd1 gdT

Result: Cell Type Specific Marker Region Peakset

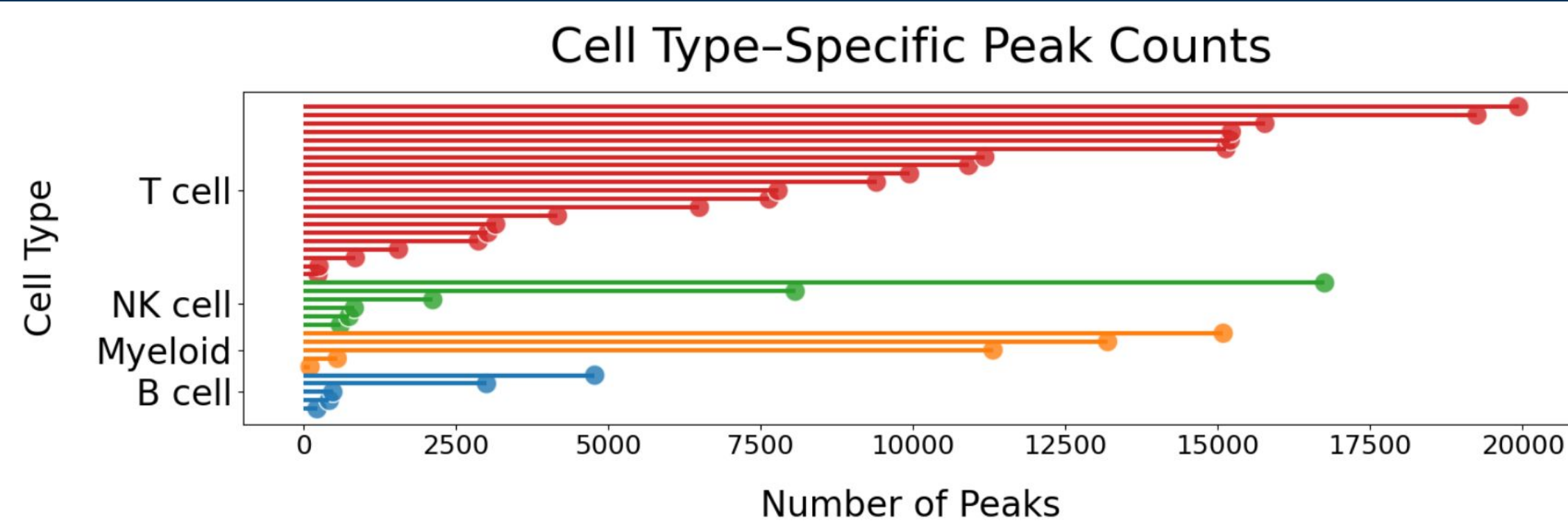
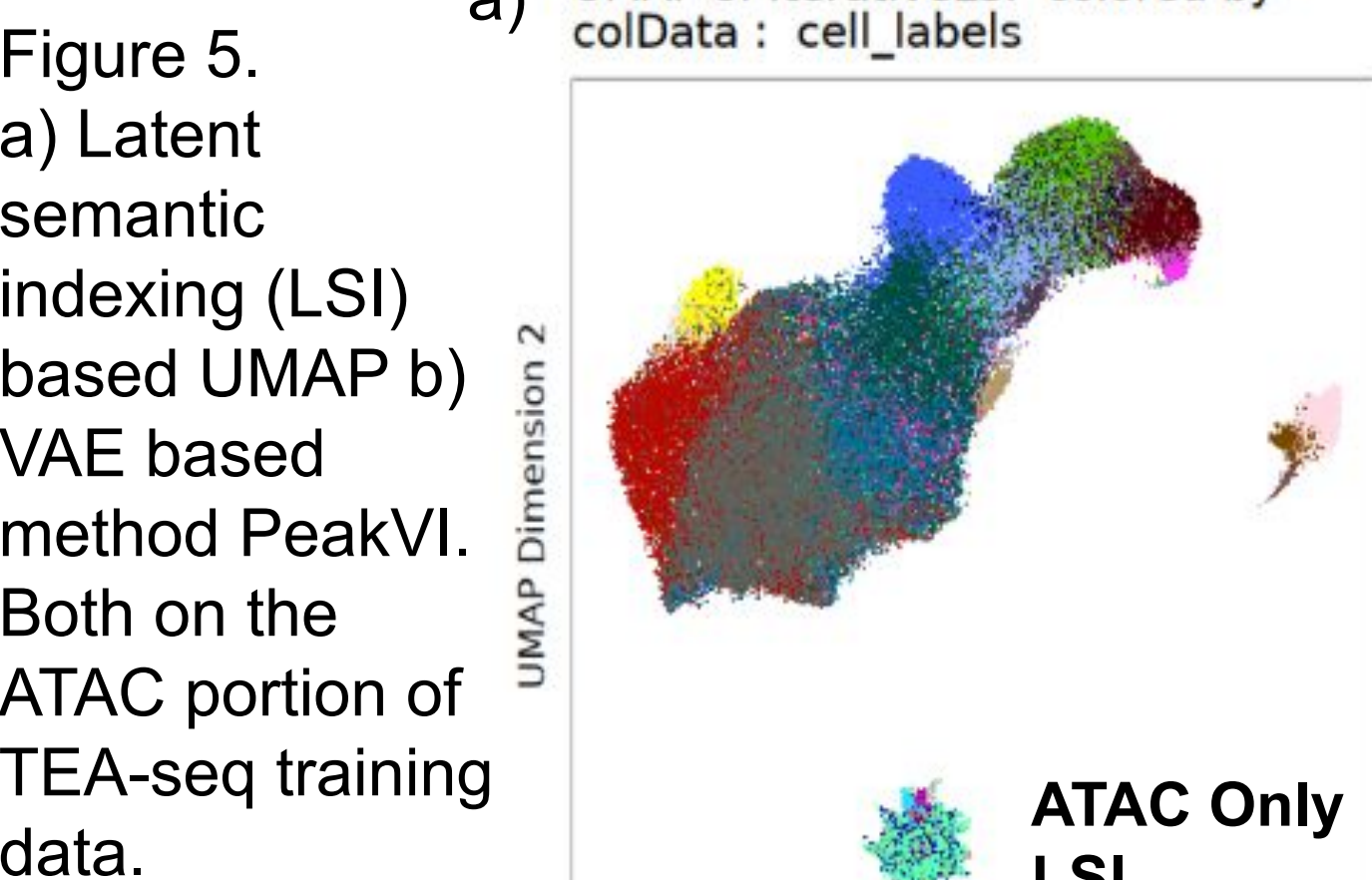


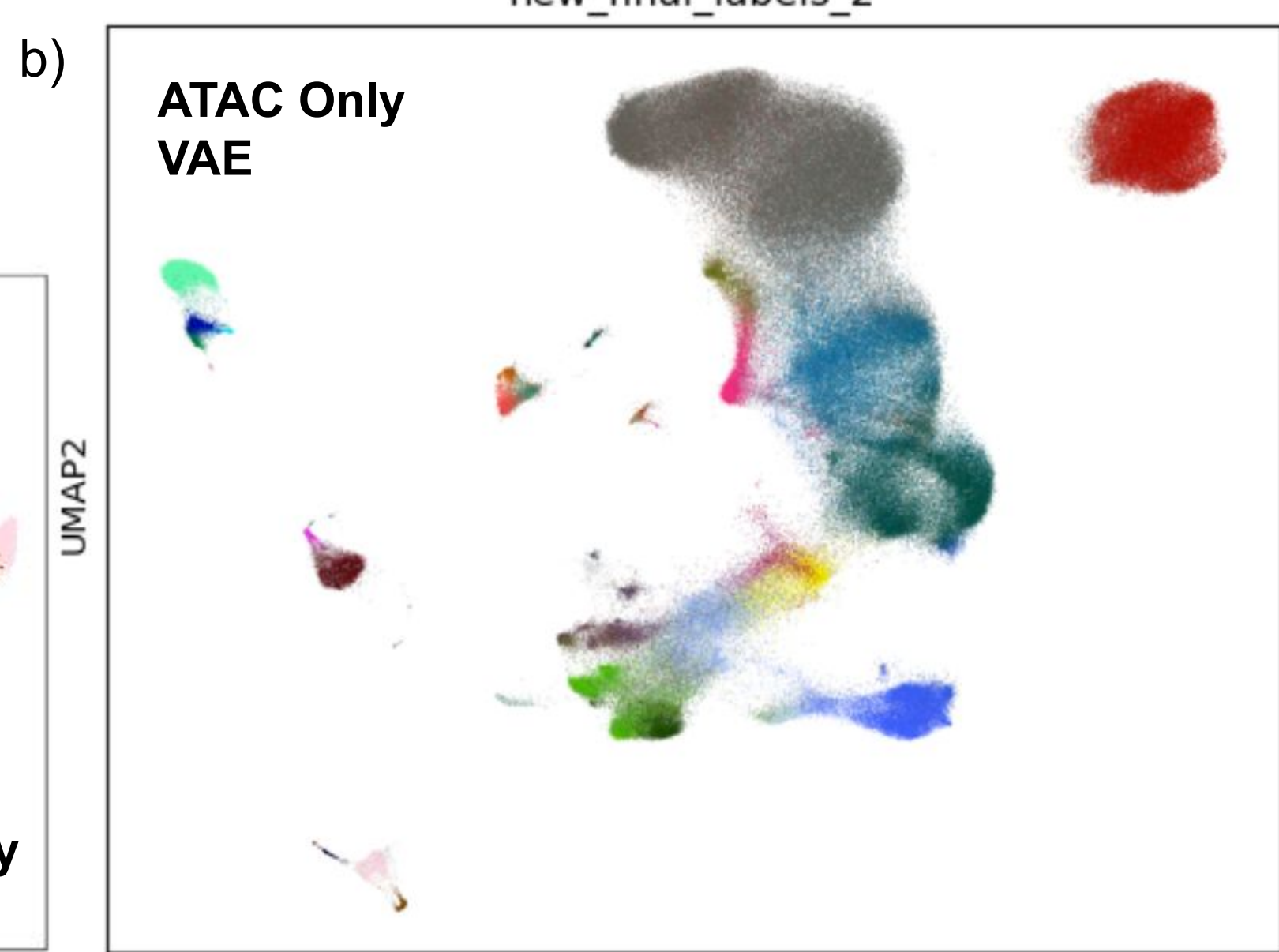
Figure 4. Number of cell type specific marker regions for each cell type.

Feature Selection Result: 75k celltype-defining marker regions out of ~500k total open regions in scATAC-seq data. Comparable to scRNA-seq feature selection ratio: 2k highly variable out of 20k total genes.

Marker regions alone do not generate accurate cell type UMAPs or clustering



- CD103+ Memory CD8 T cell
- CD27- effector B cell
- CD56bright NK cell
- CD8aa
- CD95 memory B cell
- CM CD4 T cell
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Future Directions: Insight into Adaptive NK Cell Type

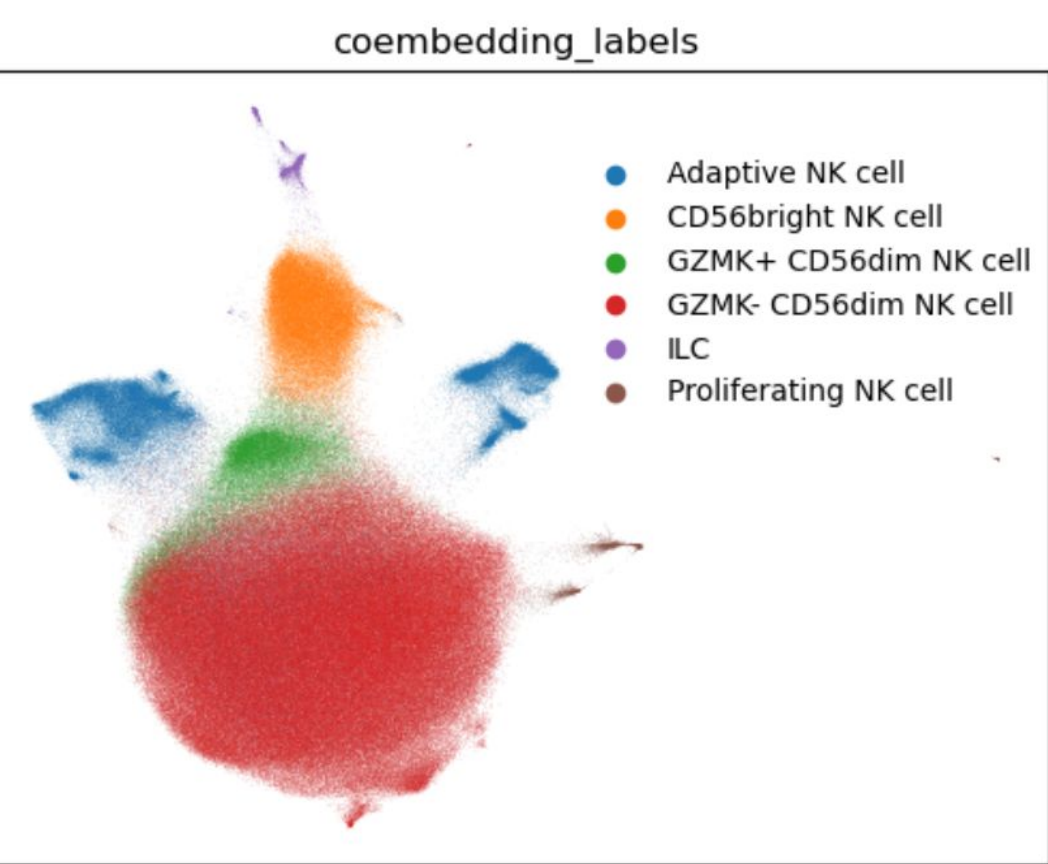


Figure 8. Cellblast co-embedding of scRNA-seq and scATAC-seq for Natural Killer (NK) cell types shows Adaptive NK clusters that were not present in TEA-seq dataset.

Utilize scATAC-seq data to explore the expansion of Adaptive NK cells in CMV positive samples.

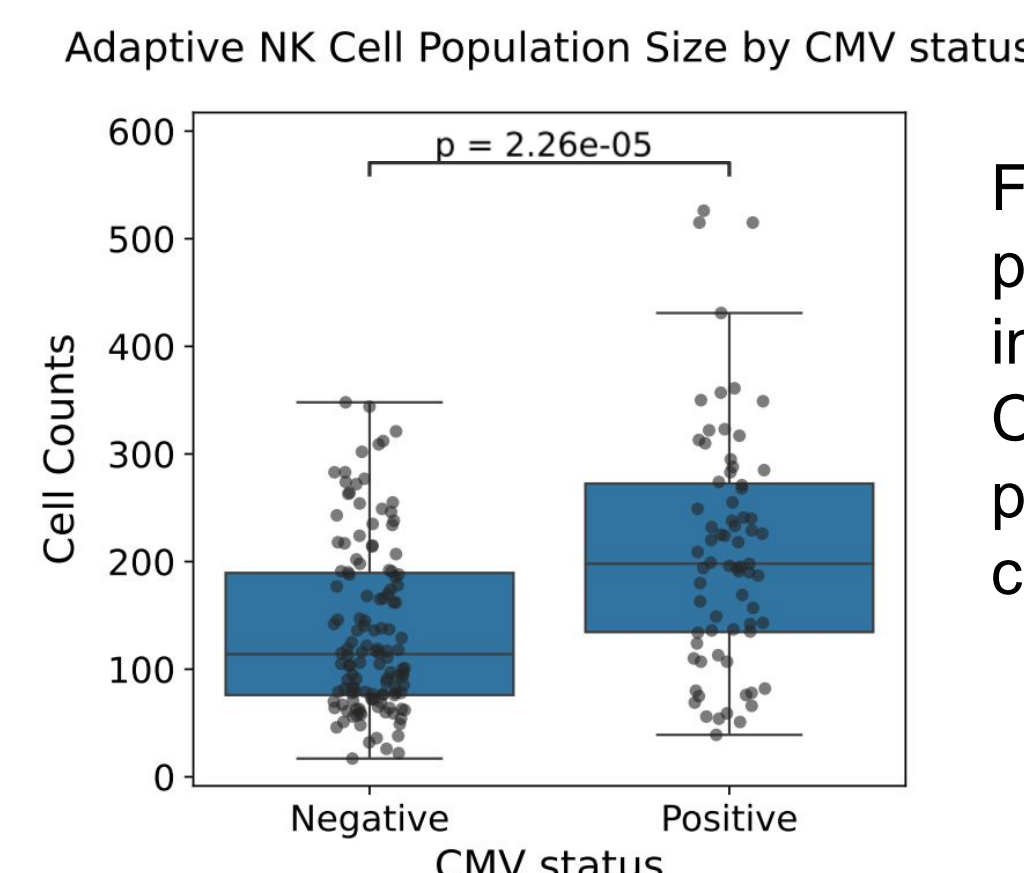


Figure 10. Adaptive NK population is significantly increased in Cytomegalovirus (CMV) positive (+) samples compared to negative (-).

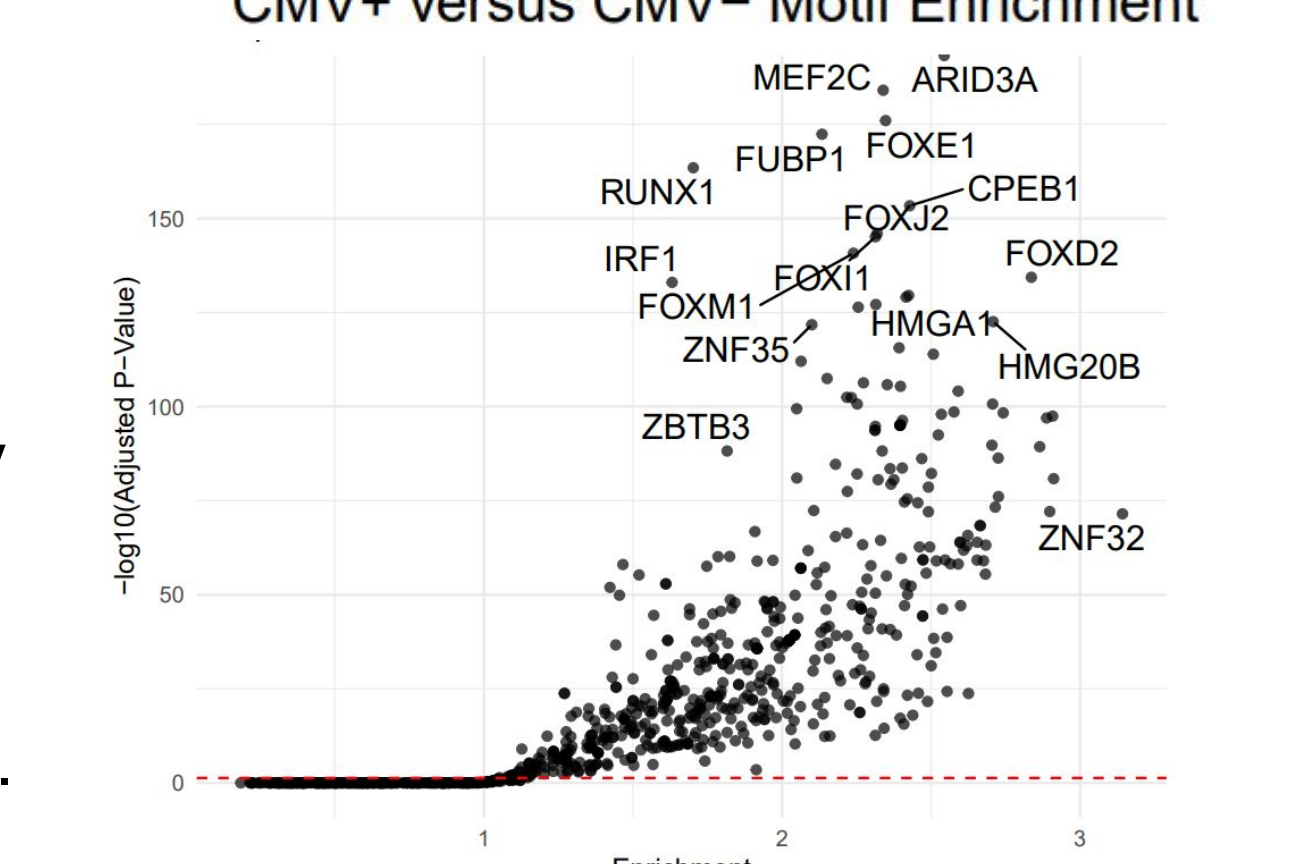
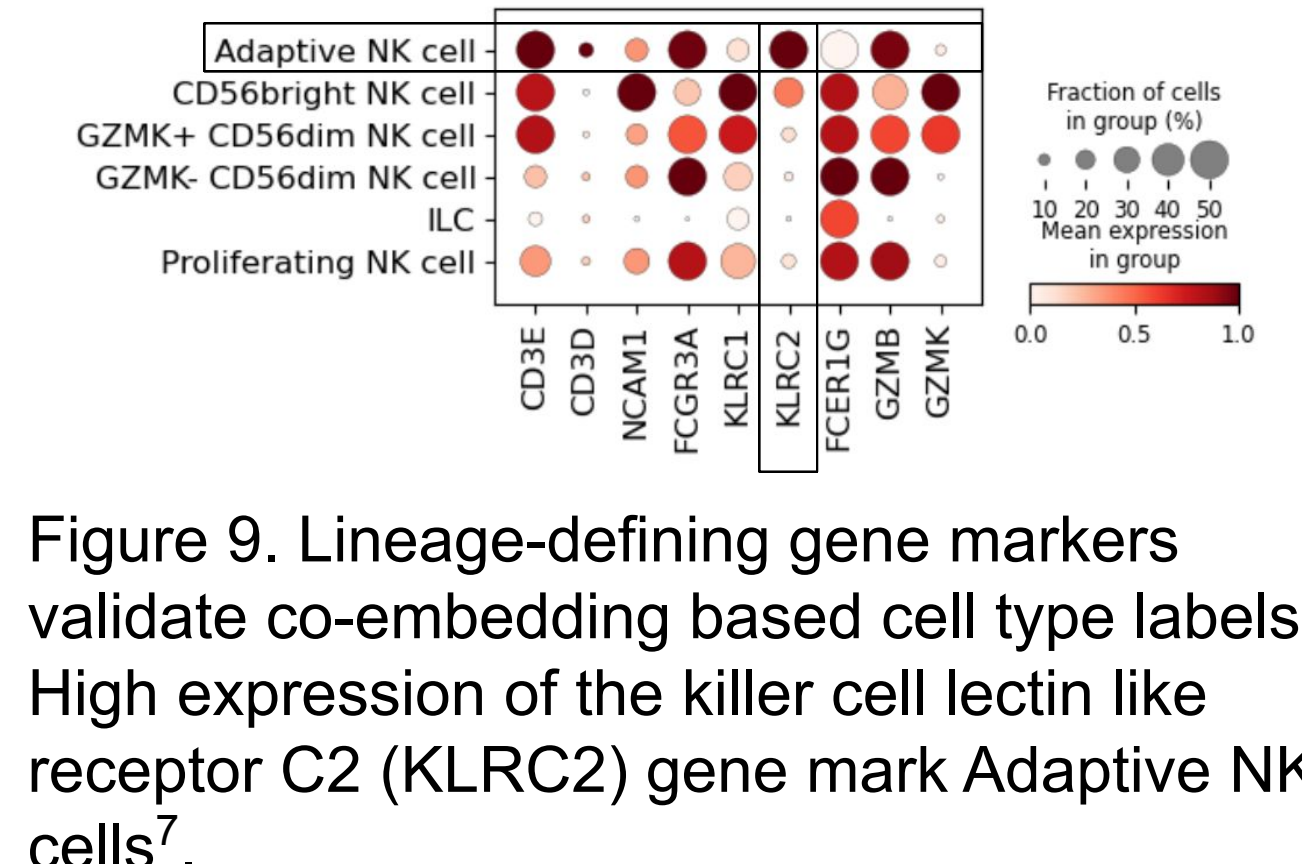


Figure 11. Using ChAI¹² & MOCHA we identify transcription factor motifs for Adaptive NK cells that are enriched in CMV+ samples.

References

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